VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Switch MPU

AFLCMC/WA

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1.0 Overview

A Switch is a Level 3 Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). Switch MPUs provide networking and connectivity between other AMS GRA elements. Switch MPUs can be housed within a Processor Chassis and Backplane (PCaB) or can be standalone units.

1.1 Purpose

The Switch MPU Volume introduces the Switch MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features (see **Table 2.1.1**) provide design guidance on what is typically expected for the Switch MPU. The interface diagrams (see **Figure 2.2.1** and **Figure 2.2.2**) contain graphical representations of the types of interfaces that can exist, from the perspective of the Switch MPU. **Table 2.3.1** contains a textual representation of the interfaces and their descriptions. **Table 3.0.1** describes what is required for the Switch MPU to achieve an AMS GRA compliance tier.

1.2 Scope

The Switch MPU volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the Switch MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	SM Compliance Documentation Template	14.0
AMS GRA Volumes	CDS MPU Volume	14.0
AMS GRA Volumes	CDS IDD	14.0

AMS GRA Models	AMS GRA Architecture Model	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of the Switch MPU.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the Switch MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	PCaB MPU Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the Switch MPU and Related Entities.

2.0 Switch

As part of the infrastructure of the AMS GRA, the Switch MPU takes on several roles related to interconnecting the network interfaces of the architecture. There are two physical configurations for the Switch MPU: PCaB and standalone. The PCaB configuration is a Line Replaceable Module (LRM) hosted in a PCaB. The standalone configuration interfaces to the Mission Platform and other MPUs over Mission Platform-provided physical interconnects.

2.1 Key Features

Table 2.1.1 identifies the key features of Switches and the benefit of each features. This table is not prescriptive but identifies a number of attributes that AMS GRA definitions and interfaces are designed to support. This table is included in the Switch documentation template to summarize the specific features supported by a Switch instantiation.

Key Feature	Benefit
Maintainability	Supports generation of fault logs
Support Required Network Speed/Protocols	Provides support for network protocols (e.g., Ethernet, InfiniBand)

Table 2.1.1: The Switch Key Features Table Describes the Features Supported By a Switch Instantiation.

2.2 Interface Diagram

Figure 2.2.1 and **Figure 2.2.2** depict the Switch MPU and its external interfaces for the PCaB configuration and Standalone configuration, respectively. Unlike the standalone configuration, Test/Debug and Platform Data in the PCaB configuration are the only interfaces that can provide a direct connection from the Switch to the platform. The PCaB and Standalone interface diagrams do not represent mutually exclusive MPU configurations. The interface diagrams only contain logical entities. Interface connections denoted as pass-through only represent situations where the sending MPU is not the original source and does not use or modify what is provided to the receiving MPU.

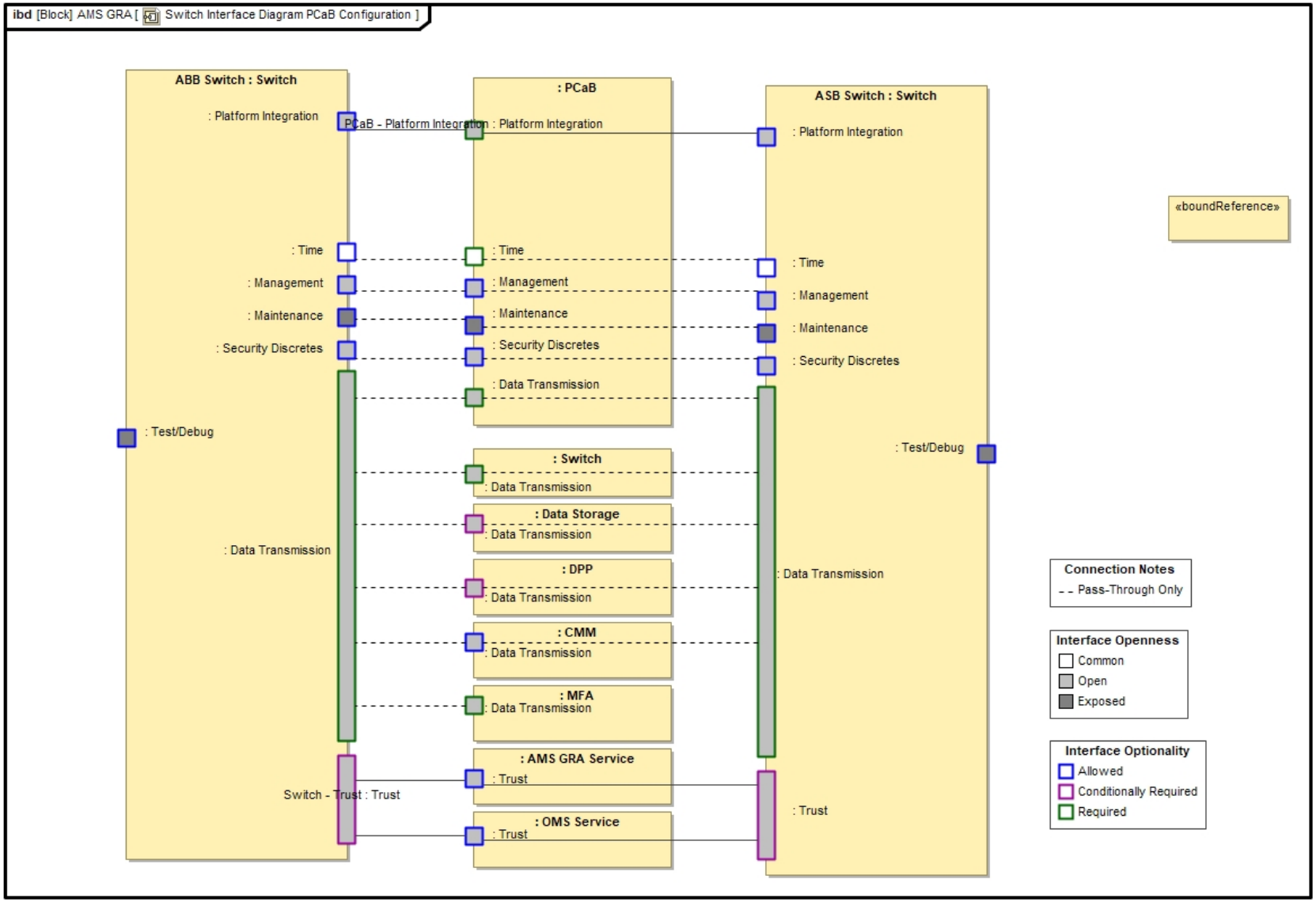


Figure 2.2.1: This Switch Interface Diagram Shows Interfaces on the Switch MPU and Their Connections When the MPU is Hosted on a PCaB.

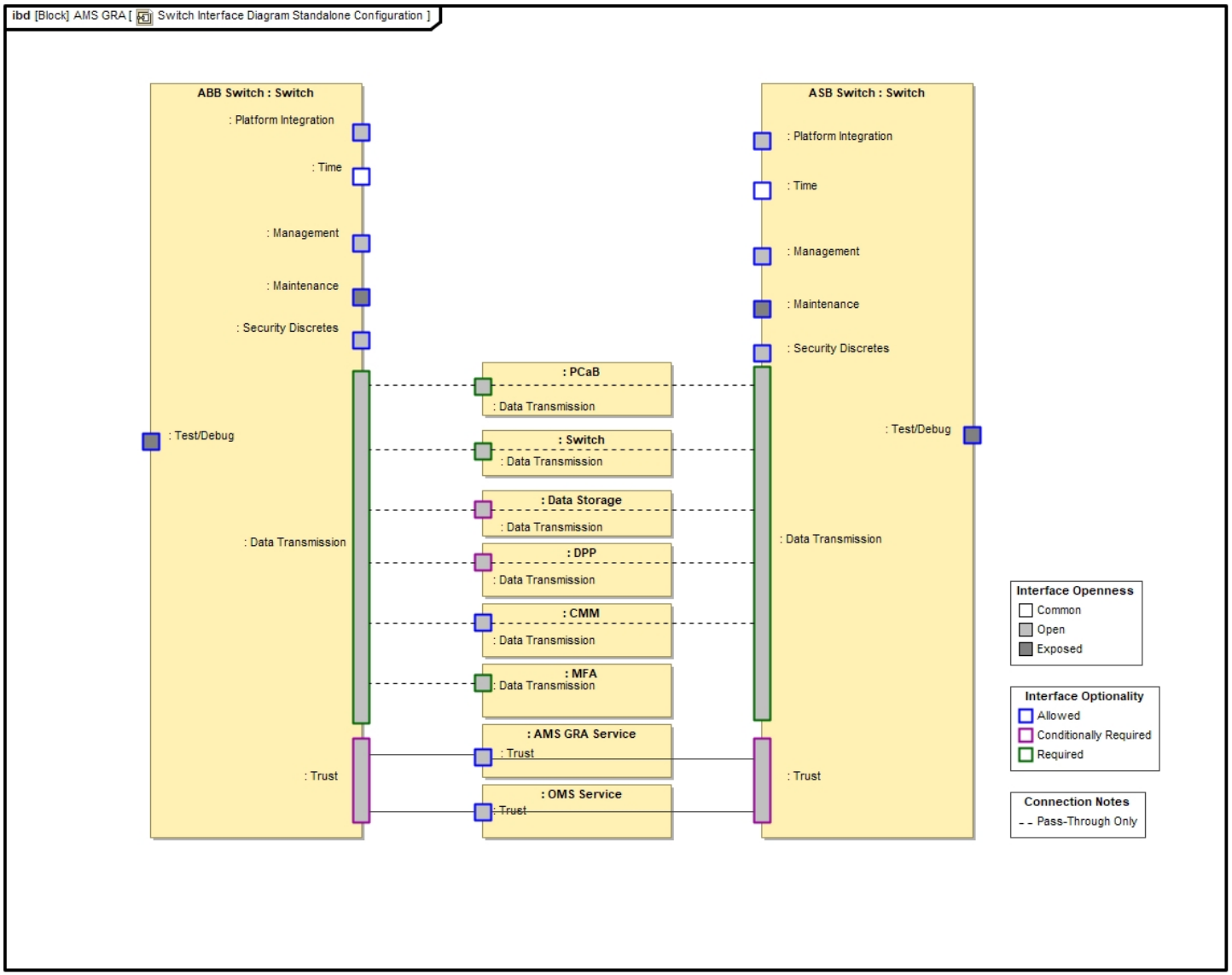


Figure 2.2.2: This Switch Interface Diagram Shows Interfaces on a Standalone Switch MPU and Their Connections. This Configuration Allows for Other Interfaces to Have Direct Connections to the Platform in Addition to Test/Debug and Platform Data.

2.3 Interface Table

Table 2.3.1 identifies the Switch interfaces depicted in **Figure 2.2.1** and **Figure 2.2.2**. When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

Interface	Documentation	Interface Optionality	Condition
Data Transmission	The Data Transmission interface represents the interface support used to facilitate a data transmission connection. The interface is not concerned with what type of data is being transmitted, it only denotes what MPUs can be connected for transmitting data.	Required	N/A
Maintenance	The Maintenance interface provides the capability to perform preventative and corrective maintenance for the Switch installed on the platform.	Allowed	N/A
Management	The Management interface provides the capability to support management of the MPU in an AMS GRA compliant mission system. An example of management functions for the Switch includes centralized control and monitoring (e.g. IPMI).	Allowed	N/A
Platform Integration	The Platform Integration interface represents the general physical interfaces that an MPU will typically have with the host platform. Examples of physical interfaces include, but are not limited to, mounting, electrical, hydraulic, and cooling connections.	Allowed	N/A
Security Discretes	The Security Discretes interface specifies the security interface between the Switch and the PCaB or platform. The interface includes items such as key loading if applicable.	Allowed	N/A
Switch Interface		Conditionally Required	
Test/Debug	The Test/Debug interface specifies an interface to use for test equipment purposes.	Allowed	N/A

Interface	Documentation	Interface Optionality	Condition
Time	The Time interface specifies the time sync interface between the Switch and other network devices.	Allowed	N/A
Trust	The Trust interface enables secure connectivity between endpoints using cryptographic means to establish a secure channel. This channel may be used to verify integrity, authenticate endpoints, or to transfer data requiring higher levels of assurance than normal data transfers, such as CPI or Software. The implementation of this interface may vary depending on assurance level, exportability, and the various mission specific requirements of the adopting program.	Conditionally Required	Required if the MPU is used to transfer mission data, software, or associated Key Material, and there are no other interfaces implemented which support integrity verification.

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 gives the applicable MPU Compliance Requirements for the Switch MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

MPU developers need to coordinate with the procuring authority and applicable MSI to receive additional guidance beyond basic compliance. The type of additional guidance the MPU developers will receive includes architectural guidance to ensure the MPU integrates within the overall Mission System architecture as well as guidance to align with Commercial-Off-The-Shelf (COTS) standards (e.g., physical form factor, cooling strategy, environmental operating conditions, etc.).

See the Mission System Volume for additional details about MPU Fault Reporting.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods In Accordance With (IAW) the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-001	An MPU shall be documented IAW the corresponding AMS GRA compliance documentation template or contract template, as applicable.	Tier 1	Inspection	
GRA-MPU-005	An MPU with CPI on a Trusted Ops platform shall cryptographically protect CPI using an open implementation.	Tier 3	Analysis	
GRA-MPU-006	An MPU that processes or decrypts CPI on a built-for-export platform shall implement the OMS Standard.	Tier 2	Analysis	
GRA-MPU-015	An MPU shall support PTP time compatible with IEEE 1588-2008	Tier 1	Inspection	MPU passes PTP packets as normal traffic
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-029	An MPU shall implement only the interfaces in the corresponding MPU interface table.	Tier 2	Inspection	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-038	An MPU shall use open, commercial standards for the Management interface.	Tier 1	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-061	An MPU shall ensure the authenticity and integrity of MPU software and firmware during MPU initialization.	Tier 1	Inspection	
GRA-MPU-062	An MPU shall support integrity verification to the Mission System over a secure communication channel.	Tier 1	Demonstration	
GRA-MPU-063	An MPU shall implement secure communications using open standards.	Tier 1	Analysis	
GRA-MPU-064	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of a software load when received over a non-secure communication channel.	Tier 1	Demonstration	
GRA-MPU-065	An MPU shall use NSA-approved algorithms to authenticate and verify the integrity of software locally stored at rest, not including bootloaders; prior to updating, loading, and executing.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-067	An MPU shall publish details of security relevant events in an open format.	Tier 1	Demonstration	
GRA-MPU-074	An MPU that processes or decrypts CPI on a built-for-export platform shall fully implement OMS Standard.	Tier 3	Analysis	
GRA-MPU-076	An MPU shall implement an optionally-enabled PTP transparent clock or PTP boundary clock functionality with software-based timestamp generation compatible with IEEE 1588-2008.	Tier 2	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUss and CDS MPUss report faults as defined in the CU IDD and CDS IDD respectively. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-087	An MPU shall implement an optionally-enabled PTP transparent clock or PTP boundary clock functionality with hardware-based timestamp generation compatible with IEEE 1588-2008.	Tier 3	Inspection	
GRA-MPU-088	An MPU shall report health and status on its Management Interface.	Tier 1	Demonstration	
GRA-MPU-091	An MPU shall report BIT information on its Management Interface.	Tier 1	Inspection	

Table 3.0.1: A Switch's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

AMS	Agile Mission Suite
BIT	Built-In Test
CDS	Cross Domain Solution
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DK	Development Kit
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
IEEE	The Institute of Electrical and Electronics Engineers
IPMI	Intelligent Platform Management Interface
LRM	Line Replaceable Module
MBSE	Model-Based Systems Engineering
MEL	Multi-Function Aperture (MFA) Encapsulation Layer
MFA	Multi-Function Aperture
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
PTP	Precision Time Protocol
RoT	Root of Trust
SAST	Static Application Security Testing

SK	Starter Kit
SM	Security Module
UCI	Universal Command and Control Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24, 2025	SK13.0
B	This revision did not introduce any significant prose changes. Adds new requirements GRA-MPU-088 and GRA-MPU-091.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Switch MPU Volume Between Documentation Releases.